

## 1.1 CP148x and CP348x

### 1.1.1 General information

Based on state-of-the-art Intel Celeron processor technology, the X20 CPUs cover a wide spectrum of demands. They can be implemented in solutions ranging from standard applications to those requiring the highest levels of performance.

The x86 100 MHz-compatible CP1483 is the entry-level X20 CPU. With an optimum price/performance ratio, it has the same basic features as all of the larger CPUs.

USB and Ethernet are included in every CPU. In addition, every CPU has a POWERLINK connection for real-time communication.

In addition, there are up to three multi-purpose slots for additional interface modules.

- Intel Celeron 650/400/266 Performance with additional I/O processor
- Entry-level CPU is Intel x86 100 MHz-compatible with additional I/O processor
- Ethernet, POWERLINK V1/V2 and USB onboard
- Modular expansion of interfaces
- Compact Flash as removable application memory
- Fan-free or exchangeable fan
- Extremely compact

## 1.1.2 Order data



CP1483, CP1484, CP1485, CP1486 <sup>1)</sup>



CP3484, CP3485, CP3486 <sup>1)</sup>

| Model number | Short description                                                                                                                                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X20CP1483    | X20 CPU, x86 100 MHz Intel compatible, 32 MB DRAM, 128 KB SRAM, exchangeable application memory: CompactFlash, 1 insert slot for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately. |
| X20CP1483-1  | X20 CPU, x86 100 MHz Intel compatible, 64 MB DRAM, 128 KB SRAM, exchangeable application memory: CompactFlash, 1 insert slot for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately. |
| X20CP1484    | X20 CPU, Celeron 266-compatible, 32 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 1 insert slot for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately.         |
| X20CP1484-1  | X20 CPU, Celeron 266-compatible, 64 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 1 insert slot for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately.         |
| X20CP1485    | X20 CPU, Celeron 400, 32 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 1 insert slot for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately.                    |
| X20CP1485-1  | X20 CPU, Celeron 400, 64 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 1 insert slot for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately.                    |
| X20CP1486    | X20 CPU, Celeron 650, 64 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 1 insert slot for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately.                    |
| X20CP3484    | X20 CPU, Celeron 266-compatible, 32 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 3 insert slots for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately.        |
| X20CP3484-1  | X20 CPU, Celeron 266-compatible, 64 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 3 insert slots for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately.        |

Table 1: X20 CPUs - Order data

| Model number                | Short description                                                                                                                                                                                                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X20CP3485                   | X20 CPU, Celeron 400, 32 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 3 insert slots for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately. |
| X20CP3485-1                 | X20 CPU, Celeron 400, 64 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 3 insert slots for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately. |
| X20CP3486                   | X20 CPU, Celeron 650, 64 MB DRAM, 1 MB SRAM, exchangeable application memory: CompactFlash, 3 insert slots for X20IF modules, 2 USB interfaces, 1 RS232 interface, 1 Ethernet interface 10/100 Base-T, 1 POWERLINK V1/V2 interface, order program memory separately. |
| <b>Program memory</b>       |                                                                                                                                                                                                                                                                      |
| 5CFCRD.0512-04              | CompactFlash 512 MB B&R                                                                                                                                                                                                                                              |
| 5CFCRD.1024-04              | CompactFlash 1024 MB B&R                                                                                                                                                                                                                                             |
| 5CFCRD.2048-04              | CompactFlash 2048 MB B&R                                                                                                                                                                                                                                             |
| 5CFCRD.4096-04              | CompactFlash 4096 MB B&R                                                                                                                                                                                                                                             |
| 5CFCRD.8192-04              | CompactFlash 8192 MB B&R                                                                                                                                                                                                                                             |
|                             |                                                                                                                                                                                                                                                                      |
| 5CFCRD.0064-03              | CompactFlash 64 MB ATA/IDE SiliconSystems                                                                                                                                                                                                                            |
| 5CFCRD.0128-03              | CompactFlash 128 MB ATA/IDE SiliconSystems                                                                                                                                                                                                                           |
| 5CFCRD.0256-03              | CompactFlash 256 MB ATA/IDE SiliconSystems                                                                                                                                                                                                                           |
| 5CFCRD.0512-03              | CompactFlash 512 MB ATA/IDE SiliconSystems                                                                                                                                                                                                                           |
| 5CFCRD.1024-03              | CompactFlash 1024 MB ATA/IDE SiliconSystems                                                                                                                                                                                                                          |
| 5CFCRD.2048-03              | CompactFlash 2048 MB ATA/IDE SiliconSystems                                                                                                                                                                                                                          |
| 5CFCRD.4096-03              | CompactFlash 4096 MB ATA/IDE SiliconSystems                                                                                                                                                                                                                          |
| 5CFCRD.8192-03              | CompactFlash 8192 MB ATA/IDE SiliconSystems                                                                                                                                                                                                                          |
| <b>Optional accessories</b> |                                                                                                                                                                                                                                                                      |
| 4A0006.00-000               | Lithium battery, 3 V / 950 mAh, button cell                                                                                                                                                                                                                          |
| 0AC201.91                   | Lithium batteries, 4 pcs., 3 V / 950 mAh, button cell                                                                                                                                                                                                                |
| X20IFxxxx                   | Communication with POWERLINK, X2X Link, CAN bus, Profibus DP, RS232, RS422, RS485 (also see Overview "Communication in the X20 IF module", on page 146)                                                                                                              |
| X20AC0EF1                   | X20 CPU exchangeable fan                                                                                                                                                                                                                                             |
| <b>Included in delivery</b> |                                                                                                                                                                                                                                                                      |
| 4A0006.00-000               | Backup battery (see also section 1.1.18 "Changing the Lithium battery" on page 21)                                                                                                                                                                                   |
| -                           | Interface module slot covers                                                                                                                                                                                                                                         |
| X20AC0SR1                   | X20 locking plate (right)                                                                                                                                                                                                                                            |
| X20TB12                     | X20 terminal block, 12-pin, 24 V coded                                                                                                                                                                                                                               |

Table 1: X20 CPUs - Order data

1) Modules CP1486 and CP3486 have exchangeable fan

**1.1.3 Technical data**

| Product ID                                                                         | CP1483                                                                                                                                  | CP1484                 | CP3484 | CP1485      | CP3485 | CP1486             | CP3486 |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------|-------------|--------|--------------------|--------|
| Short description                                                                  |                                                                                                                                         |                        |        |             |        |                    |        |
| System module                                                                      | CPU                                                                                                                                     |                        |        |             |        |                    |        |
| Processor                                                                          | x86 100 comp.                                                                                                                           | Celeron 266 comp.      |        | Celeron 400 |        | Celeron 650        |        |
| Interfaces                                                                         | 1x RS232, 1x Ethernet, 1x POWERLINK V1/V2, 2x USB, 1x X2X Link                                                                          |                        |        |             |        |                    |        |
| Controller                                                                         |                                                                                                                                         |                        |        |             |        |                    |        |
| Fastest task class cycle time                                                      | 1 ms                                                                                                                                    | 800 µs                 |        | 400 µs      |        | 200 µs             |        |
| Typical instruction cycle time                                                     | 0.076 µs                                                                                                                                | 0.022 µs               |        | 0.015 µs    |        | 0.01 µs            |        |
| Data and program code L1 cache                                                     | 16 KB                                                                                                                                   | 2 x 16 KB              |        |             |        |                    |        |
| L2 cache                                                                           | -                                                                                                                                       |                        |        | 256 KB      |        |                    |        |
| Standard memory<br>Working memory (SDRAM)<br>CPxxxx<br>CPxxxx-1<br>User RAM (SRAM) | 32 MB<br>64 MB<br>128 KB                                                                                                                | 32 MB<br>64 MB<br>1 MB |        |             |        | 64 MB<br>-<br>1 MB |        |
| Remanent variables                                                                 | 32 KB                                                                                                                                   | 64 KB                  |        | 256 KB      |        |                    |        |
| FPU                                                                                | Yes                                                                                                                                     |                        |        |             |        |                    |        |
| Integrated I/O processor                                                           | Processes I/O data points in the background                                                                                             |                        |        |             |        |                    |        |
| Data buffering<br>Lithium battery<br>Battery monitoring                            | At least 3 years<br>Yes                                                                                                                 |                        |        |             |        |                    |        |
| CompactFlash slot                                                                  | 1                                                                                                                                       |                        |        |             |        |                    |        |
| Real-time clock                                                                    | Nonvolatile memory, resolution 1 second                                                                                                 |                        |        |             |        |                    |        |
| Modular interface slots                                                            | 1                                                                                                                                       | 1                      | 3      | 1           | 3      | 1                  | 3      |
| Interfaces                                                                         |                                                                                                                                         |                        |        |             |        |                    |        |
| Interface IF1<br>Type<br>Design<br>Maximum transfer rate                           | RS232<br>Contact via 12-pin terminal block TB12<br>115.2 kBit/s                                                                         |                        |        |             |        |                    |        |
| Interface IF2<br>Type<br>Design<br>Transfer rate<br>Cable length                   | Ethernet<br>Shielded RJ45 port<br>10/100 MBit/s<br>Max. 100 m between two stations (segment length)                                     |                        |        |             |        |                    |        |
| Interface IF3<br>Fieldbus<br>Type<br>Design<br>Transfer rate<br>Cable length       | POWERLINK V1/V2<br>100 Base-T (ANSI/IEEE 802.3)<br>Shielded RJ45 port<br>100 MBit/s<br>Max. 100 m between two stations (segment length) |                        |        |             |        |                    |        |
| Interfaces IF4 and IF5                                                             | USB Rev. 1.1                                                                                                                            |                        |        |             |        |                    |        |
| IF6 interface                                                                      | X2X Link                                                                                                                                |                        |        |             |        |                    |        |

Table 2: X20 CPUs - Technical data

| Product ID                                                                                          | CP1483                                                                                                                                                           | CP1484 | CP3484 | CP1485 | CP3485 | CP1486                                                                                                                             | CP3486 |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|------------------------------------------------------------------------------------------------------------------------------------|--------|
| CPU and X2X Link supply                                                                             |                                                                                                                                                                  |        |        |        |        |                                                                                                                                    |        |
| Input voltage                                                                                       | 24 VDC (-15% / +20%)                                                                                                                                             |        |        |        |        |                                                                                                                                    |        |
| Input current                                                                                       | Max. 2.2 A                                                                                                                                                       |        |        |        |        |                                                                                                                                    |        |
| Reverse polarity protection                                                                         | Yes                                                                                                                                                              |        |        |        |        |                                                                                                                                    |        |
| Fuse                                                                                                | Integrated, cannot be exchanged                                                                                                                                  |        |        |        |        |                                                                                                                                    |        |
| X2X Link supply output                                                                              |                                                                                                                                                                  |        |        |        |        |                                                                                                                                    |        |
| Rated output power                                                                                  | 7.0 W                                                                                                                                                            |        |        |        |        |                                                                                                                                    |        |
| Parallel operation                                                                                  | Yes <sup>1)</sup>                                                                                                                                                |        |        |        |        |                                                                                                                                    |        |
| Redundant operation of X2X Link supply                                                              | Yes                                                                                                                                                              |        |        |        |        |                                                                                                                                    |        |
| I/O supply input                                                                                    |                                                                                                                                                                  |        |        |        |        |                                                                                                                                    |        |
| Input voltage                                                                                       | 24 VDC (-15% / +20%)                                                                                                                                             |        |        |        |        |                                                                                                                                    |        |
| Fuse                                                                                                | Recommended pre-fusing max. 10 A slow-blow                                                                                                                       |        |        |        |        |                                                                                                                                    |        |
| I/O supply output                                                                                   |                                                                                                                                                                  |        |        |        |        |                                                                                                                                    |        |
| Rated output voltage                                                                                | 24 VDC                                                                                                                                                           |        |        |        |        |                                                                                                                                    |        |
| Permitted contact load                                                                              | 10.0 A                                                                                                                                                           |        |        |        |        |                                                                                                                                    |        |
| General supply                                                                                      |                                                                                                                                                                  |        |        |        |        |                                                                                                                                    |        |
| Status indicators                                                                                   | Overload, operating status, module status, RS232 data transfer                                                                                                   |        |        |        |        |                                                                                                                                    |        |
| Diagnostics<br>Module run/error<br>Overload<br>RS232 data transfer                                  | Yes, with status LED and software status<br>Yes, with status LED and software status<br>Yes, with status LED                                                     |        |        |        |        |                                                                                                                                    |        |
| Electrical isolation<br>X2X bus supply<br>I/O supply                                                | Yes<br>No                                                                                                                                                        |        |        |        |        |                                                                                                                                    |        |
| General information                                                                                 |                                                                                                                                                                  |        |        |        |        |                                                                                                                                    |        |
| Status indicators                                                                                   | CPU function, overtemperature, Ethernet, POWERLINK, CompactFlash, battery                                                                                        |        |        |        |        |                                                                                                                                    |        |
| Diagnostics<br>CPU function<br>Over-temperature<br>Ethernet<br>POWERLINK<br>CompactFlash<br>Battery | Yes, with status LED<br>Yes, with status LED<br>Yes, with status LED<br>Yes, with status LED<br>Yes, with status LED<br>Yes, with status LED and software status |        |        |        |        |                                                                                                                                    |        |
| Fan diagnostics                                                                                     | -                                                                                                                                                                |        |        |        |        | Yes, with software status                                                                                                          |        |
| Visual Components capability                                                                        | Yes                                                                                                                                                              |        |        |        |        |                                                                                                                                    |        |
| ACOPOS capability                                                                                   | Yes                                                                                                                                                              |        |        |        |        |                                                                                                                                    |        |
| Cooling                                                                                             | Fan-free                                                                                                                                                         |        |        |        |        | Fan-free with derating (see operating conditions, on page 6)<br>Exchangeable fans for entire temperature range<br>Fan is monitored |        |

Table 2: X20 CPUs - Technical data (cont.)

## CPUx CP148x and CP348x

| Product ID                                                                                                                | CP1483                                                             | CP1484 | CP3484 | CP1485 | CP3485 | CP1486                                                                      | CP3486 |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------|--------|--------|--------|-----------------------------------------------------------------------------|--------|
| Electrical isolation<br>PLC - IF1/IF4/IF5<br>PLC - IF2/IF3/IF6<br>IF1/IF4/IF5 - IF2/IF3/IF6<br>IF1 - IF4/IF5<br>IF4 - IF5 | No<br>Yes<br>Yes<br>No<br>No                                       |        |        |        |        |                                                                             |        |
| Power consumption - without memory card, interface module, or USB                                                         | 6.0 W                                                              | 10.5 W | 10.5 W | 10.5 W | 10.5 W | 13.5 W                                                                      | 13.5 W |
| Internal power consumption of the X2X Link and I/O supply <sup>2)</sup><br>Bus<br>I/O internal                            | 1.42 W<br>0.6 W                                                    |        |        |        |        |                                                                             |        |
| Certification                                                                                                             | CE, C-UL-US <sup>3)</sup> , GOST-R                                 |        |        |        |        |                                                                             |        |
| <b>Operating conditions</b>                                                                                               |                                                                    |        |        |        |        |                                                                             |        |
| Operating temperature<br>Horizontal installation<br><br>Vertical installation                                             | 0 to +55°C<br><br>0 to +50°C                                       |        |        |        |        | 0 to +55°C<br>Fan-free: 0 to +45°C<br>0 to +55°C<br>Fan-free is not allowed |        |
| Relative humidity                                                                                                         | 5 - 95%, non-condensing                                            |        |        |        |        |                                                                             |        |
| Mounting orientation                                                                                                      | Horizontal or vertical                                             |        |        |        |        |                                                                             |        |
| Installation above sea level<br>0 - 2000 m<br>>2000 m                                                                     | No derating<br>Reduction of ambient temperature by 0.5°C per 100 m |        |        |        |        |                                                                             |        |
| Protection type                                                                                                           | IP20                                                               |        |        |        |        |                                                                             |        |
| <b>Storage and transport conditions</b>                                                                                   |                                                                    |        |        |        |        |                                                                             |        |
| Temperature                                                                                                               | -25 to +70°C                                                       |        |        |        |        |                                                                             |        |
| Relative humidity                                                                                                         | 5 - 95%, non-condensing                                            |        |        |        |        |                                                                             |        |

Table 2: X20 CPUs - Technical data (cont.)

| Product ID                        | CP1483                                                                                                                                                                                                                                               | CP1484        | CP3484        | CP1485        | CP3485        | CP1486        | CP3486        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| <b>Mechanical characteristics</b> |                                                                                                                                                                                                                                                      |               |               |               |               |               |               |
| Dimensions (W x H x D) [mm]       | 150 x 99 x 85                                                                                                                                                                                                                                        | 150 x 99 x 85 | 200 x 99 x 85 | 150 x 99 x 85 | 200 x 99 x 85 | 150 x 99 x 85 | 200 x 99 x 85 |
| Comment                           | Order application memory (CompactFlash) separately<br>Backup battery included in delivery<br>X20 locking plate (right) included in delivery<br>X20 terminal block (12-pin) included in delivery<br>Interface module slot covers included in delivery |               |               |               |               |               |               |

Table 2: X20 CPUs - Technical data (cont.)

- 1) In parallel operation, only 75 % of the rated power can be assumed. Please ensure that all parallel operating power supplies are switched on and off simultaneously.
- 2) The specified values are maximum values. The exact calculation is described in chapter 3 "Mechanical and electrical configuration", section 22 "Power dissipation and supply modules", on page 129. The calculation is also available for download as a data sheet with the other module documentation on the B&R homepage.
- 3) CP1483: C-UL-US listing in preparation

#### 1.1.4 X20 CPUs - Status LEDs

| LED   | Color     | Status   | Description                                                                                                               |
|-------|-----------|----------|---------------------------------------------------------------------------------------------------------------------------|
| R/E   | Green     | On       | Application running                                                                                                       |
|       | Red       | On       | SERVICE mode                                                                                                              |
| RDY/F | Yellow    | On       | SERVICE or BOOT mode                                                                                                      |
|       | Red       | On       | Over-temperature                                                                                                          |
| S/E   | Green/red |          | Status/error LED. The status of the LED is described in section "S/E LED".                                                |
| EPL   | Green     | On       | A link to the POWERLINK remote station has been established.                                                              |
|       |           | Blinking | A link to the POWERLINK remote station has been established. The LED blinks when Ethernet activity is present on the bus. |
| ETH   | Green     | On       | A link to the Ethernet remote station has been established.                                                               |
|       |           | Blinking | A link to the Ethernet remote station has been established. The LED blinks when Ethernet activity is present on the bus.  |
| CF    | Yellow    | On       | CompactFlash OK                                                                                                           |
|       | Green     | On       | CompactFlash active                                                                                                       |
| DC OK | Yellow    | On       | CPU power supply OK                                                                                                       |
|       | Red       | On       | Backup battery is empty                                                                                                   |

Table 3: X20 CPUs - CPU status indicator

**S/E LED**

The status/error LED is a green/red dual LED. The status LEDs can have different meanings depending on operating mode.

Ethernet TCP/IP mode

With Automation Studio Version V 2.5.3 and later, the interface can be operated as a purely Ethernet TCP/IP interface.

| Green - status | Description                                                                 |
|----------------|-----------------------------------------------------------------------------|
| On             | The POWERLINK interface is operated purely as an Ethernet TCP/IP interface. |

Table 4: X20 CPUs - Status/error LED - Ethernet TCP/IP operating mode

POWERLINK V1

| Status LED           |          | Status of the POWERLINK station                                                                                                                                                                                                   |
|----------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Green                | Red      |                                                                                                                                                                                                                                   |
| On                   | Off      | The POWERLINK station is running with no errors.                                                                                                                                                                                  |
| Off                  | On       | A fatal system error has occurred. The error type can be read using the PLC logbook. It concerns an irreparable problem. The system cannot properly carry out its tasks. This status can only be changed by resetting the module. |
| Blinking alternately |          | The POWERLINK managing node failed. This error code can only occur in controlled node operation. This means that the set station number lies within the range \$01 - \$FD.                                                        |
| Off                  | Blinking | System failure. The red blinking LED signals an error code (see section "System failure error codes" on page 10).                                                                                                                 |

Table 5: X20 CPUs - Status/error LED - POWERLINK V1 operating mode

POWERLINK V2

| Red - error | Description                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| On          | The POWERLINK interface has encountered an error (failed Ethernet frames, increased number of collisions on the network, etc.). |

Table 6: X20 CPUs - Status/error LED as error LED - POWERLINK V2 operating mode

| Green - status                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Off<br>NOT_ACTIVE                                  | <p><u>Managing Node (MN)</u><br/>The bus is monitored for POWERLINK frames. If a frame is not received within the configured time window (timeout), the interface goes directly into PRE_OPERATIONAL_1 status (single flash).<br/>If, however, POWERLINK communication is detected before this time passes, the interface goes directly into the BASIC_ETHERNET status (flickering).</p> <p><u>Controlled Node (CN)</u><br/>The bus is monitored for POWERLINK frames. If a frame is not received within the configured time window (timeout), the interface goes directly into BASIC_ETHERNET status (flickering). If, however, POWERLINK communication is detected during this time, the interface goes directly into the PRE_OPERATIONAL_1 status (single flash).</p> |
| Flickering green (approx. 10 Hz)<br>BASIC_ETHERNET | <p>The interface is in BASIC_ETHERNET status, and is operated purely as an Ethernet TCP/IP interface.</p> <p><u>Managing Node (MN)</u><br/>This status can only be changed by resetting the interface.</p> <p><u>Controlled Node (CN)</u><br/>If POWERLINK communication is detected while in this status, the interface goes into the PRE_OPERATIONAL_1 state (single flash).</p>                                                                                                                                                                                                                                                                                                                                                                                       |
| Single flash (approx. 1 Hz)<br>PRE_OPERATIONAL_1   | <p>The interface status is PRE_OPERATIONAL_1.</p> <p><u>Managing Node (MN)</u><br/>The MN starts the operation of the "reduced cycle". Collisions are allowed on the bus. There is not yet any cyclic communication.</p> <p><u>Controlled Node (CN)</u><br/>The CN waits until it receives an SoC frame and then switches to PRE_OPERATIONAL_2 status (double flash).</p>                                                                                                                                                                                                                                                                                                                                                                                                |
| Double flash (approx. 1 Hz)<br>PRE_OPERATIONAL_2   | <p>The interface status is PRE_OPERATIONAL_2.</p> <p><u>Managing Node (MN)</u><br/>The MN begins with the cyclic communication (cyclic input data is not yet evaluated). The CNs are configured in this status.</p> <p><u>Controlled Node (CN)</u><br/>In this status, the interface is normally configured by the manager. After this, a command changes the status to READY_TO_OPERATE (triple flash).</p>                                                                                                                                                                                                                                                                                                                                                             |
| Triple flash (approx. 1 Hz)<br>READY_TO_OPERATE    | <p>The interface status is READY_TO_OPERATE.</p> <p><u>Managing Node (MN)</u><br/>Cyclic and asynchronous communication. Received PDO data is ignored.</p> <p><u>Controlled Node (CN)</u><br/>The configuration of the interface is complete. Normal cyclic and asynchronous communication. The PDO data sent corresponds to the PDO mapping used. However, cyclic data is not yet evaluated.</p>                                                                                                                                                                                                                                                                                                                                                                        |
| On<br>OPERATIONAL                                  | The interface status is OPERATIONAL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 7: X20 CPUs - Status/error LED as status LED - POWERLINK V2 operating mode

| Green - status                       | Description                                                                                                                                                                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Blinking (approx. 2.5 Hz)<br>STOPPED | <p>The interface status is STOPPED.</p> <p><u>Managing Node (MN)</u><br/>This status is not possible for the MN.</p> <p><u>Controlled Node (CN)</u><br/>No output data is produced and no input data is received. Only the appropriate command from the manager can enter or leave this state.</p> |

Table 7: X20 CPUs - Status/error LED as status LED - POWERLINK V2 operating mode (cont.)

## System failure error codes

Incorrect configuration or defective hardware can cause a system failure error.

The error is displayed via the red error LED using four switch-on phases. The switch-on phases are either 150 ms or 600 ms long. Error code outputs are repeated cyclically after 2 seconds.

Legend:

- ... 150 ms
- ... 600 ms
- Pause ... 2 s delay

| Error description | Error code displayed by red status LED |   |   |   |       |   |   |   |   |       |
|-------------------|----------------------------------------|---|---|---|-------|---|---|---|---|-------|
| RAM Error         | •                                      | • | • | – | Pause | • | • | • | – | Pause |
| Hardware Errors   | –                                      | • | • | – | Pause | – | • | • | – | Pause |

Table 8: X20 CPUs - Status/error LED as error LED - System failure error codes

### 1.1.5 Status LEDs for integrated power supply

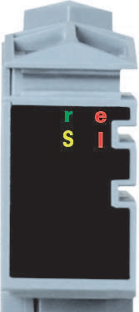
| Figure                                                                            | LED   | Color                           | Status       | Description                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|-------|---------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | r     | Green                           | Off          | Module supply not connected                                                                                                                                                                                           |
|                                                                                   |       |                                 | Single flash | Reset mode                                                                                                                                                                                                            |
|                                                                                   |       |                                 | Blinking     | Preoperational mode                                                                                                                                                                                                   |
|                                                                                   |       |                                 | On           | RUN mode                                                                                                                                                                                                              |
|                                                                                   | e     | Red                             | Off          | Module supply not connected or everything is OK                                                                                                                                                                       |
|                                                                                   |       |                                 | Double flash | Indicates one of the following conditions: <ul style="list-style-type: none"> <li>• X2X Link power supply is overloaded</li> <li>• I/O supply too low</li> <li>• Input voltage for X2X Link supply too low</li> </ul> |
|                                                                                   | e + r | Steady red / single green flash |              | Invalid firmware                                                                                                                                                                                                      |
|                                                                                   | S     | Yellow                          | Off          | No RS232 activity                                                                                                                                                                                                     |
|                                                                                   |       |                                 | On           | The LED is on, when data is being sent or received via the RS232 interface                                                                                                                                            |
|                                                                                   | I     | Red                             | Off          | X2X Link supply in the acceptable range                                                                                                                                                                               |
|                                                                                   |       |                                 | On           | X2X Link power supply is overloaded                                                                                                                                                                                   |

Table 9: X20 CPUs - Status indicators for integrated power supply

## 1.1.6 Operating and connection elements

### CP148x

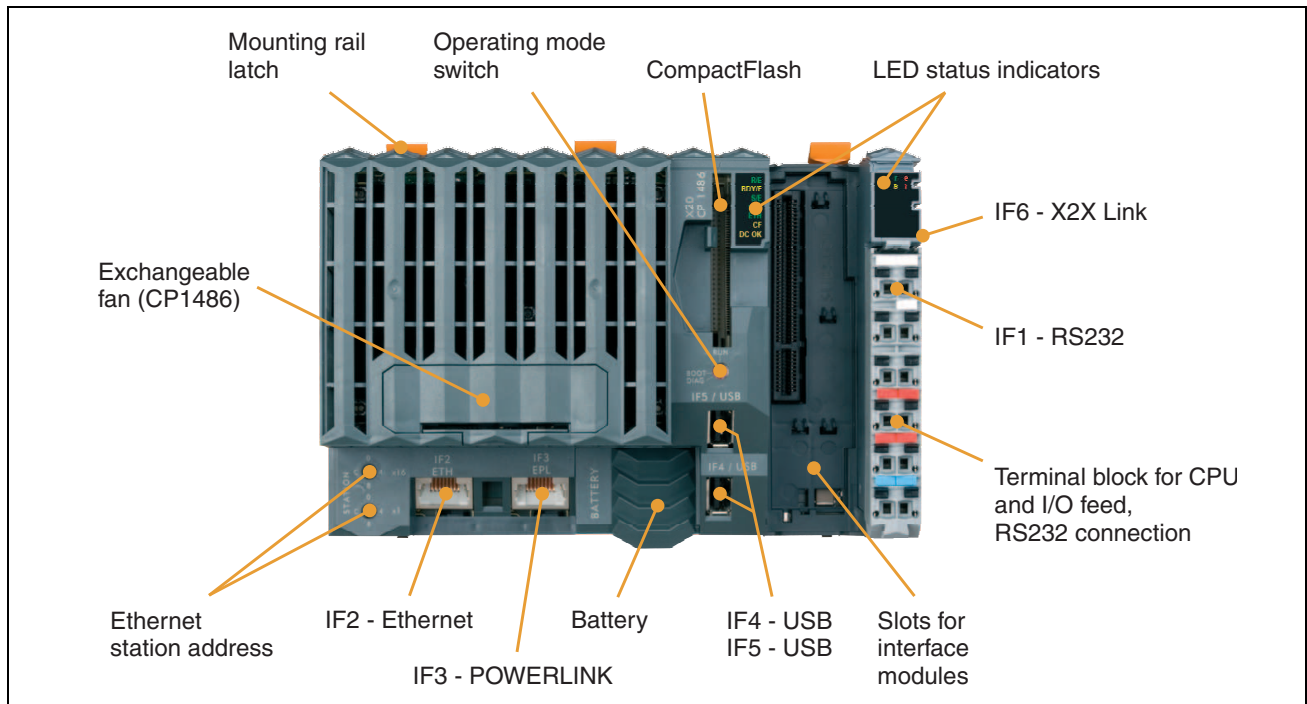


Figure 1: X20 CPUs - Operating elements for CP1483, CP1485, and CP1486

### CP348x

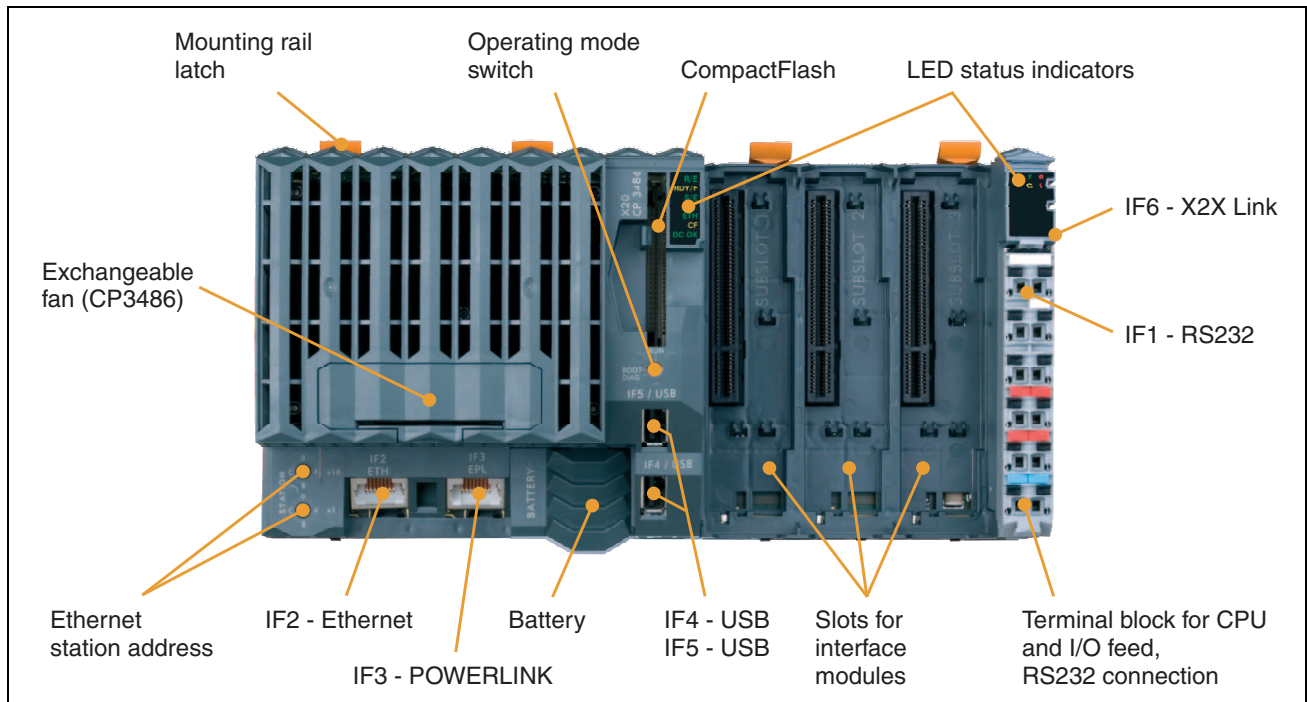


Figure 2: X20 CPUs - Operating elements for CP3484, CP3485, and CP3486

### 1.1.7 Slot for application memory

Program memory is required to operate the CPUs. The program memory is CompactFlash. It is not included with the delivery of the CPUs, instead it must be ordered as an accessory.

## Information:

The CompactFlash card cannot be removed during operation.

### 1.1.8 Operating mode switch

An operating mode switch is used to set the operating mode.



Figure 3: X20 CPUs - Operating mode switch

| Switch position | Operating mode | Description                                                                                                                                                                                                                  |
|-----------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BOOT            | Boot           | In this switch position the default B&R Automation Runtime (AR) is started, and the runtime system can be installed using the online interface (B&R Automation Studio). The User Flash is deleted after the download begins. |
| RUN             | Run            | RUN mode                                                                                                                                                                                                                     |
| DIAG            | Diagnostics    | The CPU boots in diagnostics mode. Program sections in User RAM and User FlashPROM are not initialized. After Diagnostics mode, the CPU always boots with a <b>warm restart</b> .                                            |

Table 10: X20 CPUs - Operating modes

### 1.1.9 CPU supply

A power supply comes integrated in the X20 CPUs. It is equipped with a feed for the CPU, the X2X Link, and the internal I/O supply. The feed to the CPU/X2X Link supply is electrically isolated.

Redundancy of the X2X Link supply possible by parallel operation of multiple supply modules.

#### Pin assignments

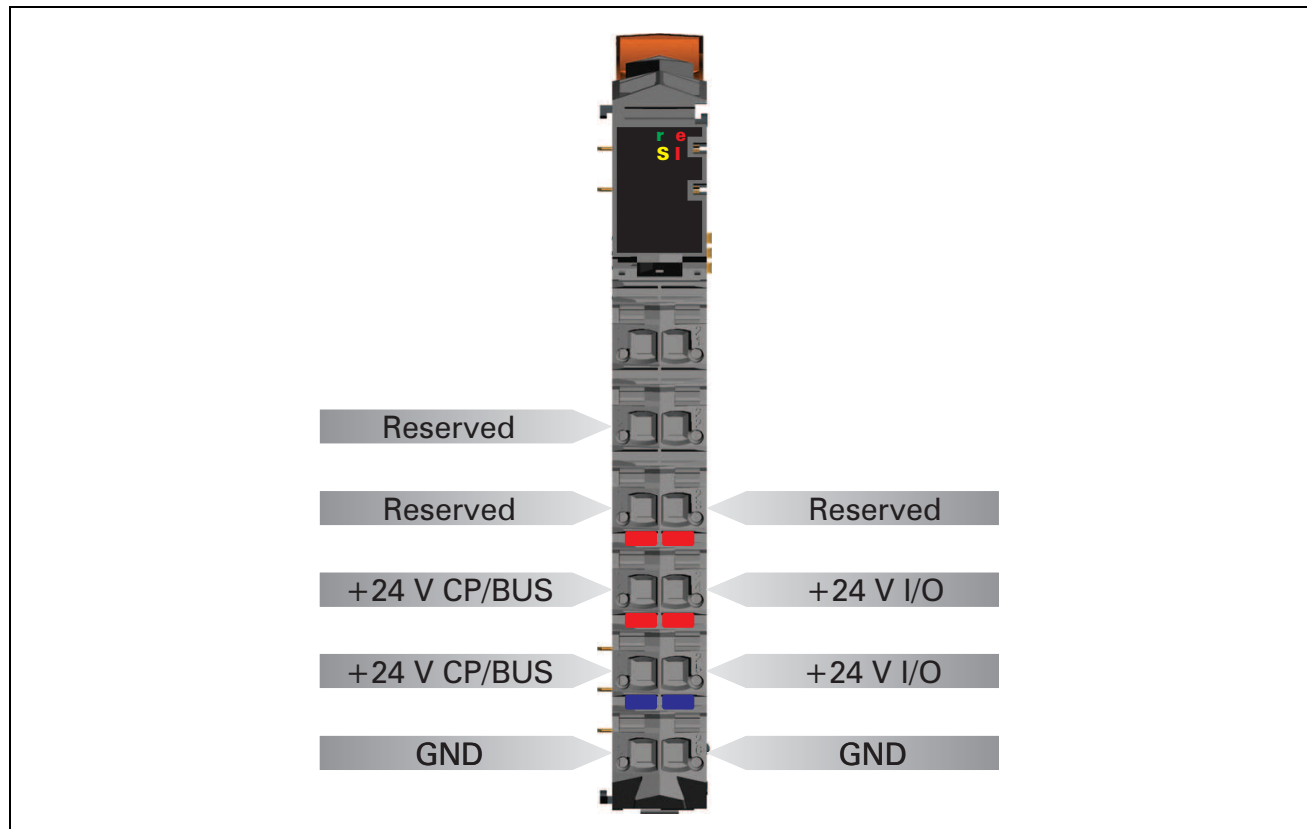


Figure 4: X20 CPUs - Pin assignments - Integrated power supply

## Connection examples

### With two separate supplies

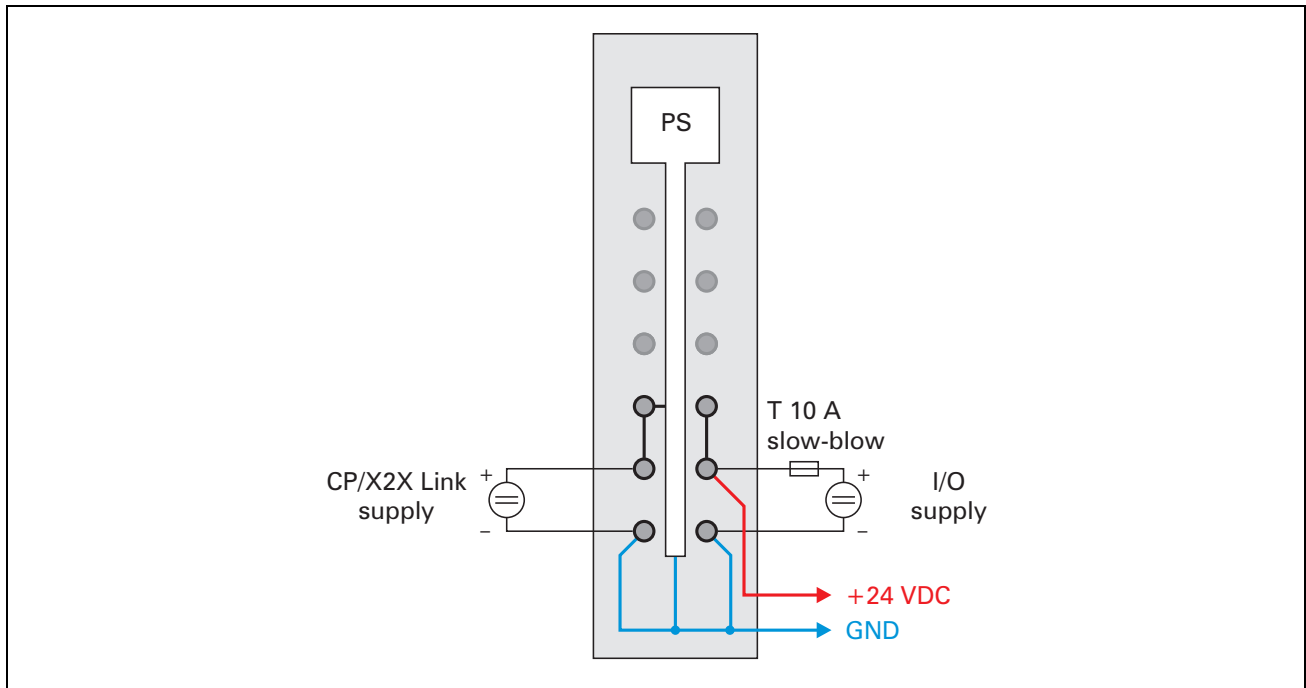


Figure 5: X20 CPUs - Connection example with two separate supplies

### With a supply and jumper

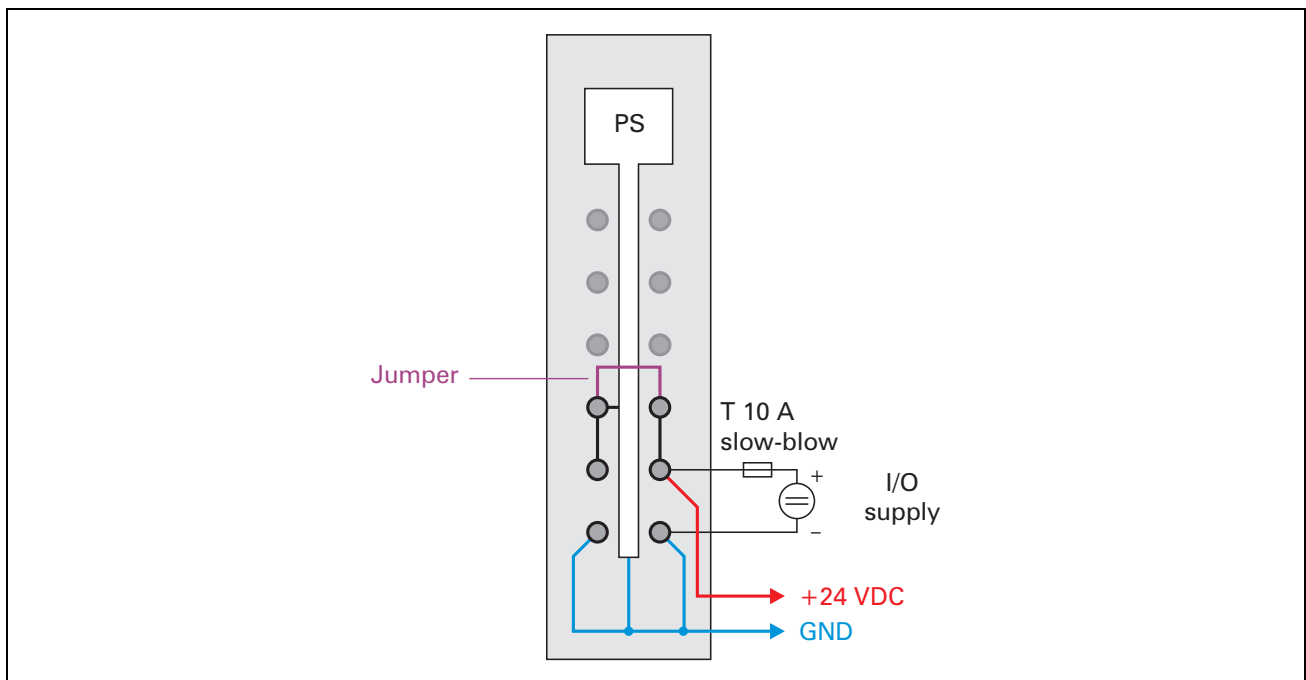


Figure 6: X20 CPUs - Connection example with a supply and jumper

### **1.1.10 RS232 interface (IF1)**

The RS232 interface is not electrically isolated. It can be used as an online interface for communicating with the programming device.

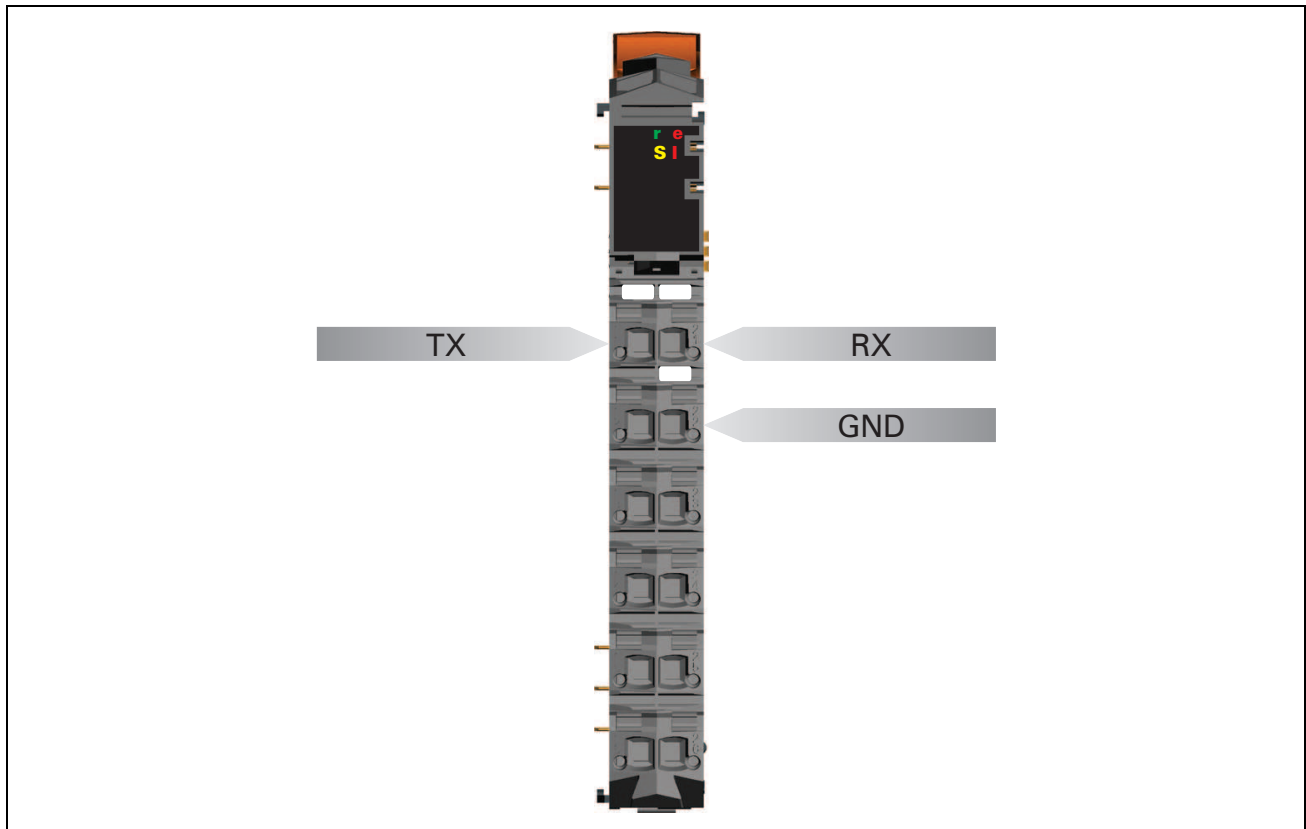


Figure 7: X20 CPUs - Pin assignments - RS232 interface (IF1)

### 1.1.11 Ethernet interface (IF2)

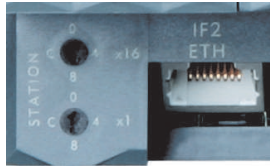


Figure 8: X20 CPUs - Ethernet interface (IF2)

IF2 is an Ethernet interface. The connection is made using a 10/100 BASE-T Twisted Pair RJ45 socket.

The INA2000 station number for the Ethernet interface is set with both hex switches.

Please note the following section: 9.1.2 "Cabling guidelines for X20 bus controllers with Ethernet cable" on page 265

## Information:

The Ethernet interface (IF2) is not suited for POWERLINK (see interface IF3 on page 18).

### Pin assignments

| Pin | Assignment  |                |
|-----|-------------|----------------|
| 1   | RxD         | Receive data   |
| 2   | RxD\        | Receive data\  |
| 3   | TxD         | Transmit data  |
| 4   | Termination |                |
| 5   | Termination |                |
| 6   | TxD\        | Transmit data\ |
| 7   | Termination |                |
| 8   | Termination |                |

Table 11: X20 CPUs - Pin assignments - Ethernet interface (IF2)

**1.1.12 POWERLINK interface (IF3)****POWERLINK V1**

Station numbers are permitted between \$00 and \$FD. The station number can be set using software.

| Switch position | Description                                                         |
|-----------------|---------------------------------------------------------------------|
| \$00            | Operation as managing node.                                         |
| \$01 - \$FD     | Station number for POWERLINK station. Operation as controlled node. |
| \$FE - \$FF     | Reserved, switch position is not permitted.                         |

Table 12: X20 CPUs - Station number POWERLINK V1

**POWERLINK V2**

Station numbers are permitted between \$01 and \$F0. The station number can be set using software.

| Switch position | Description                                                         |
|-----------------|---------------------------------------------------------------------|
| \$00            | Reserved, switch position is not permitted.                         |
| \$01 - \$EF     | Station number for POWERLINK station. Operation as controlled node. |
| \$F0            | Operation as managing node.                                         |
| \$F1 - \$FF     | Reserved, switch position is not permitted.                         |

Table 13: X20 CPUs - Station number POWERLINK V2

**Ethernet TCP/IP mode**

Starting with Automation Studio Version V 2.5.3 and with Automation Runtime V 2.90, the interface can be operated purely as an Ethernet TCP/IP interface.

The INA2000 station number can be set using the B&R Automation Studio software.

## Pin assignments

Please note the following section: 9.1.2 "Cabling guidelines for X20 bus controllers with Ethernet cable" on page 265

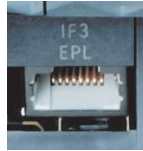


Figure 9: X20 CPUs - POWERLINK interface (IF3)

| Pin | Assignment  |                |
|-----|-------------|----------------|
| 1   | RxD         | Receive data   |
| 2   | RxD\        | Receive data\  |
| 3   | TxD         | Transmit data  |
| 4   | Termination |                |
| 5   | Termination |                |
| 6   | TxD\        | Transmit data\ |
| 7   | Termination |                |
| 8   | Termination |                |

Table 14: X20 CPUs - Pin assignments for POWERLINK interface (IF3)

### 1.1.13 USB interfaces (IF4 and IF5)



Figure 10: X20 CPUs - USB interfaces (IF4 and IF5)

IF4 and IF5 are USB interfaces. The connection is made using a USB 1.1 interface.

The USB interfaces can only be used for devices which have been released by B&R (e.g. floppy disk drive, DiskOnKey or dongle).

## Information:

The USB interfaces cannot be used as online communication interfaces.

### 1.1.14 Slots for interface modules

The CPU's are equipped with one or three slots for interface modules.

The X20 System can be connected to various bus or network systems by selecting the appropriate interface modules.

An overview of the interface modules is provided on page 390.

### 1.1.15 Exchangeable fan

The X20 CPU's CP1486 and CP3486 are delivered with exchangeable fans. Therefore, they can be used throughout the full temperature range from 0 - 55°C. A fan is not necessary up to 45°C.

A replacement fan can be ordered using the order number X20AC0EF1.

#### Changing the fan

- 1) Press in fan lock with thumb and pull out fan.
- 2) Insert new fan into CPU until the lock clicks into place.

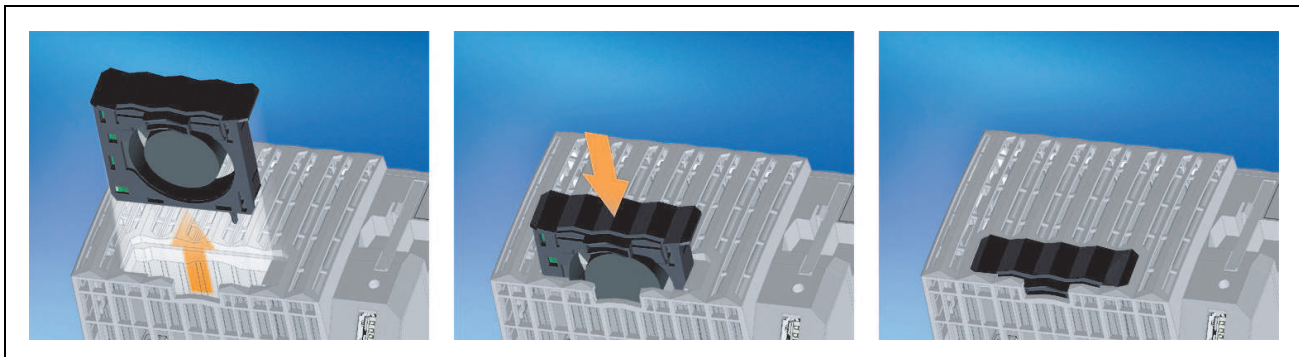


Figure 11: X20 CPU's - Tool-free fan replacement

### 1.1.16 Overtemperature shut-off

To prevent damage, a shut-off/reset is triggered when the CPU reaches 100°C.

The following errors are entered in the log book:

| Error numbers | Error description                                   |
|---------------|-----------------------------------------------------|
| 9204          | WARNING: System halted because of temperature check |
| 9210          | WARNING: Boot by watchdog or manual reset           |

Table 15: X20 CPU's - Log book entries for overtemperature shut-off

### 1.1.17 Data / real-time buffering

The CPUs are equipped with a backup battery. The following areas are buffered:

- Remanent variables
- User RAM
- System RAM
- Real-time clock

### Battery monitoring

The battery voltage is checked cyclically. The cyclic load test of the battery does not considerably shorten the battery life, instead it gives an early warning of weakened buffer capacity.

The status information "Battery OK" is available from the system library function "BatteryInfo" and the CPU's I/O mapping.

### Battery change interval

The battery should be changed every 4 years. Change intervals are recommended by B&R and refer to average life span and operating conditions. It is not the maximum buffer duration.

### 1.1.18 Changing the Lithium battery

The CPUs are equipped with a lithium battery. The lithium battery is placed in a separate compartment and protected by a cover.

### Backup battery data

|                                           |                                             |
|-------------------------------------------|---------------------------------------------|
| Model number<br>4A0006.00-000<br>0AC201.9 | 1 pcs.<br>5 pcs.                            |
| Short description                         | Lithium battery, 3 V / 950 mAh, button cell |
| Storage temperature                       | -20 to +60°C                                |
| Storage time                              | Max. 3 years at 30°C                        |
| Relative humidity                         | 0 - 95%, non-condensing                     |

Table 16: X20 CPUs - backup battery data

The product design allows the battery to be changed with the PLC switched either on or off. In some countries, safety regulations do not allow batteries to be changed while the module is switched on. To prevent data loss, the battery must be changed within 1 min. with the power off.

### Procedure for changing the battery

- 1) Touch the mounting rail or ground connection (not the power supply!) in order to discharge any electrostatic charge from your body.
- 2) Remove the cover for the lithium battery. To do this, slide the cover down from the CPU.

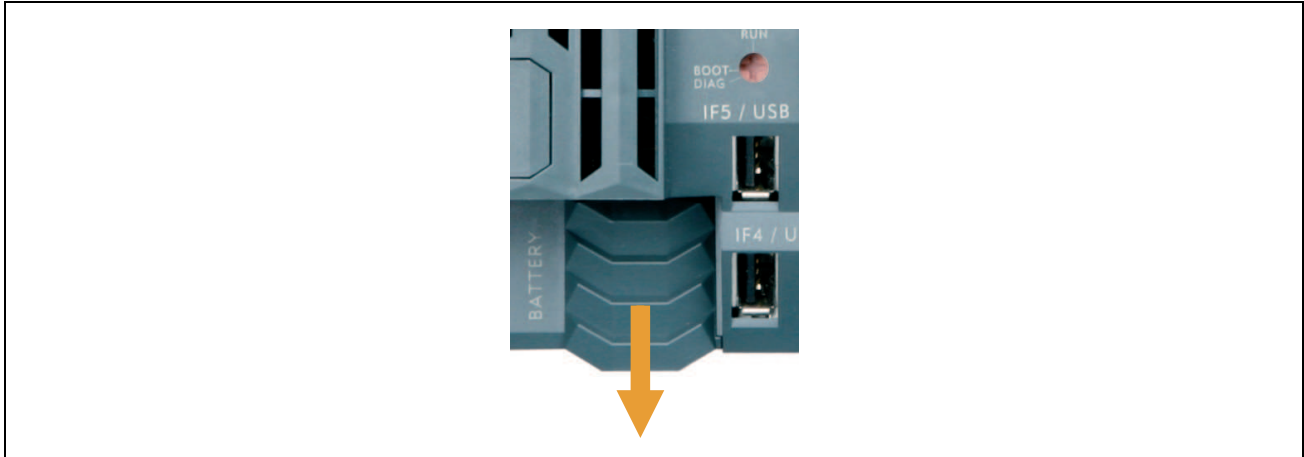


Figure 12: X20 CPUs - Removing the cover for the lithium battery

- 3) Remove the battery from the holder (do not use pliers or uninsulated tweezers -> risk of short circuiting). The battery should not be held by its edges. **Insulated** tweezers may also be used for removing the battery.

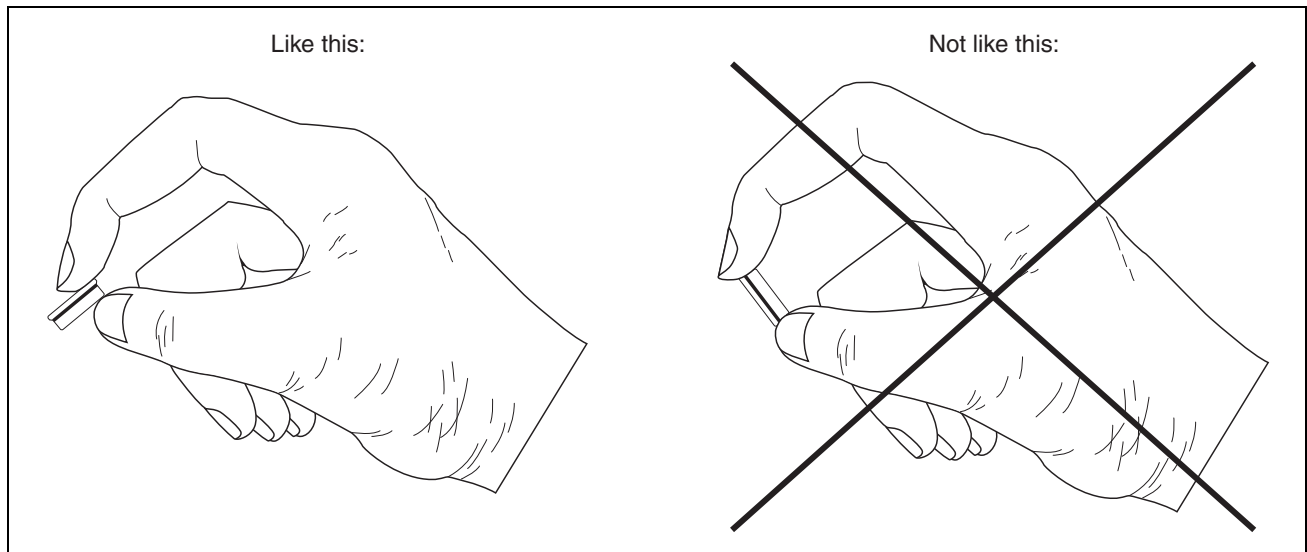


Figure 13: X20 CPUs - Correct grip for the battery

- 4) Insert the new battery with correct polarity. To do this, the battery is laid with the "+" side up on the right part of the battery holder under the USB interface IF4. Then secure the battery in the holder by pressing above the left part of the battery holder.
- 5) Replace cover.

## Information:

**Lithium batteries are considered hazardous waste. Used batteries should be disposed of accordingly.**

### 1.1.19 Programming the System Flash

#### General information

The CPUs are delivered with a default B&R Automation Runtime (with limited functions) already installed. This runtime system is started in Boot mode (operating mode switch position BOOT or no CompactFlash / invalid CompactFlash inserted). It initializes and operates the serial RS232 onboard interface, allowing a runtime system download via INA2000 protocol.

This runtime system download is carried out during the commissioning of the CPU. The runtime system is stored in the program memory (CompactFlash) of the processor. The runtime system can be updated later by performing a runtime system update or using the B&R Automation Studio PVI Transfer Tool.

X20 System CPUs are supported with B&R Automation Studio V 2.5.2.7 or later.

#### Runtime system download

When installing the runtime system (runtime system download) the following procedure must be carried out:

- 1) Insert CompactFlash and switch on the power to the PLC. If the switch is in BOOT position, or if a new or invalid CompactFlash card is inserted, the CPU is now in BOOT mode.
- 2) Establish an online connection between the programming device (PC or Industrial PC) and the CPU. A runtime system download is only possible using the serial RS232 onboard interface.
- 3) Start B&R Automation Studio.
- 4) Start the download procedure by selecting the **Services** command from the **Project** menu. Select **Transfer Operating System...** from the menu shown. Now follow the instructions from B&R Automation Studio.

- 5) A dialog box is displayed for configuring the runtime system version. The runtime system version is already preselected by the user's project settings. Using the drop-down menu, the runtime system versions stored in the project can be selected. Clicking on the **Browse** button allows the selected runtime system version to be loaded from the hard drive or from the CD.

Pressing **Next >** opens a pop-up window, which allows the user to select whether the modules should be downloaded with SYSTEM ROM target memory using the following runtime system download. Otherwise, modules can also be transferred later with an application download.

Pressing **Next >** brings the user to a control box where the current settings are displayed.

- 6) The download procedure is started by pressing **Finish**. Download progress is shown in a message window.

### Information:

**User flash is cleared.**

- 7) To operate the runtime system, the operating mode switch must be turned to 4 when the download procedure is completed. Reset the CPU or turn off and then on again if necessary.
- 8) The PLC is now ready for use.

### Runtime system update

When updating the runtime system (online runtime system update), the following procedure must be carried out:

- 1) An online runtime system update is only possible if a valid operating system is located on the CompactFlash.
- 2) Establish an online connection between the programming device (PC or Industrial PC) and the CPU. An online runtime update is now possible using the interfaces configured as an online interface.  
The runtime system update can be completed using either a serial online connection (INA2000 protocol), a CAN network, or an ETHERNET network, depending on the system configuration.
- 3) Start B&R Automation Studio.
- 4) Start the update procedure by calling the **Services** command from the **Project** menu. Select **Transfer Operating System...** from the menu shown. Now follow the instructions from B&R Automation Studio.

- 5) A dialog box is displayed for configuring the runtime system version. The runtime system version is already preselected by the user's project settings. Using the drop-down menu, the runtime system versions stored in the project can be selected. Clicking on the **Browse** button allows the selected runtime system version to be loaded from the hard drive or from the CD.

Pressing **Next >** opens a pop-up window, which allows the user to select whether the modules should also be downloaded with SYSTEM ROM as the target memory during the following runtime system update. Otherwise, modules can also be transferred later with an application download.

Pressing **Next >** brings the user to a control box where the current settings are displayed.

- 6) The update procedure is started by pressing **Finish**. The update progress is shown in a message box.

## Information:

**User flash is cleared.**

- 7) When the update procedure is complete, the online connection is automatically reestablished.
- 8) The PLC is now ready for use.

### Runtime system generation/update with the PVI Transfer Tool

- 1) In the **Tools** menu, select the command **Generate transfer list**.
- 2) In the dialog box that appears, select **Generate complete transfer list** and activate the option **Include operating system**. Continue by clicking on **OK**.
- 3) In the **Tools** menu, select the command **PVI Transfer Tool**. This starts the PVI Transfer Tool. For further steps, please refer to the Help for the PVI Transfer Tool (Tools - Create CompactFlash).